Operations on graphs

Outline

Operations on graphs

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Operations related to $y_2 = f(x + a)$

Operations related to $y_3 = k\, f(x)$

Operations related to $y_4 = f(k\, x)$

Operations related to $y_5 = |f(x)|$

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Operations on graphs

Introduction: Operations on graphs

- Knowing the graph of a function $y = f(x)$, we want to draw, without computations, the graphs of functions obtained from f through simple operations
- These operations correspond to **translations**, **dilations** and **reflections** of the graph
- It is a fundamental tool to quickly draw graphs starting from those of the elementary functions

The six operations

Knowing the graph of a function $y = f(x)$, by means of simple geometric transformations it is possible to draw the graph of the following functions:

$$y_1 = f(x) + a, \quad a \in \mathbb{R}$$

$$y_2 = f(x + a), \quad a \in \mathbb{R}$$

$$y_3 = k\, f(x), \quad k \in \mathbb{R}$$

$$y_4 = f(k\, x), \quad k \in \mathbb{R}$$

$$y_5 = |f(x)|$$

$$y_6 = f(|x|)$$

Translations, dilations and reflections

- Therefore, starting from the elementary functions, it is possible to build, by means of these operations, a large variety of new functions.
- These operations create **translations**, **dilations** and **reflections** of the original function.

Summary

- From the graph of $y = f(x)$ we obtain the graphs of $f(x) + a$, $f(x + a)$, $k\,f(x)$, $f(kx)$, $|f(x)|$ and $f(|x|)$
- The operations produce translations, dilations and reflections
- In the next sections we study them one at a time

Operations related to $y_1 = f(x) + a$

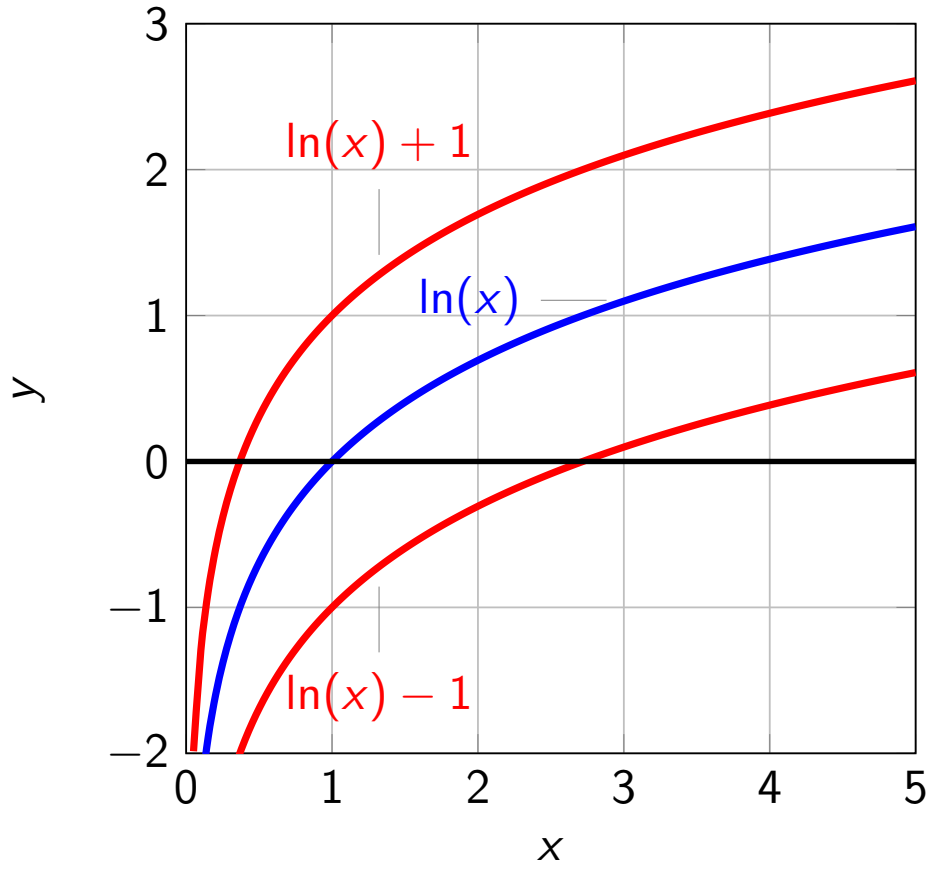
Introduction: Operations related to $y_1 = f(x) + a$

- We see what happens to the graph when we **add a constant** a to the values of the function
- All the values $f(x)$ are increased (or decreased) by the same amount: we expect a **vertical** shift of the graph

Example

Consider $y = \ln x$. Then

$$y_1 = \ln(x) + a \quad (x > 0).$$



Vertical translation

- The graph of y_1 is obtained from that of y by a **translation** of a units **upward** if $a > 0$, **downward** if $a < 0$.

Summary

$y_1 = f(x) + a$:

- Vertical translation by a units
- Upward if $a > 0$, downward if $a < 0$
- The domain does not change (e.g. $\ln(x) + a$ is defined for $x > 0$)

Operations related to $y_2 = f(x + a)$

Introduction: Operations related to $y_2 = f(x + a)$

- We see what happens to the graph when we **add a constant** a to the variable x, before applying f
- We expect a **horizontal** shift of the graph, which also changes the domain of the function

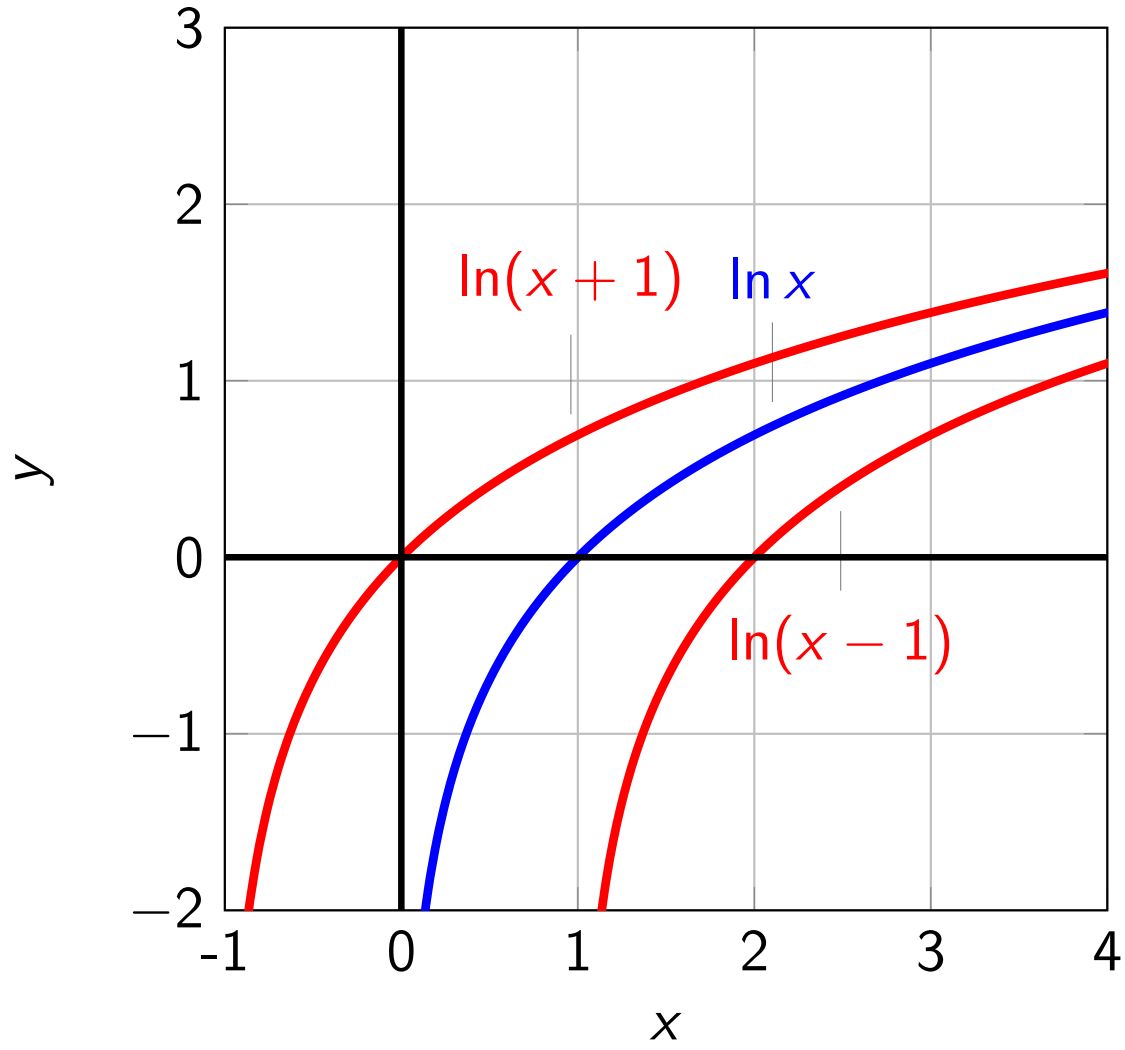
Example (1/2)

Consider again $y = \ln x$. Then

$$y_2 = \ln(x + a).$$

- Since $\ln x$ is defined for $x > 0$, $\ln(x + a)$ will be defined for $x + a > 0$, i.e., for $x > -a$.
- Since $\ln x = 0$ if $x = 1$, we have $\ln(x + a) = 0$ for $x + a = 1$, i.e., $x = 1 - a$.

Example (2/2)



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Horizontal translation

- The graph of y_2 is obtained from that of y by a **translation** of a units to the **left** if $a > 0$, to the **right** if $a < 0$.

Summary

$y_2 = f(x + a)$:

- Horizontal translation by a units
- To the left if $a > 0$, to the right if $a < 0$
- The domain shifts: $\ln(x + a)$ is defined for $x > -a$ and vanishes at $x = 1 - a$

Operations related to $y_3 = k\, f(x)$

Introduction: Operations related to $y_3 = k\, f(x)$

- We see what happens to the graph when we **multiply** the values of the function by a constant k
- We expect the graph to be "stretched" or "squeezed" in the **vertical** direction, and possibly reflected

Vertical dilation and contraction (1/2)

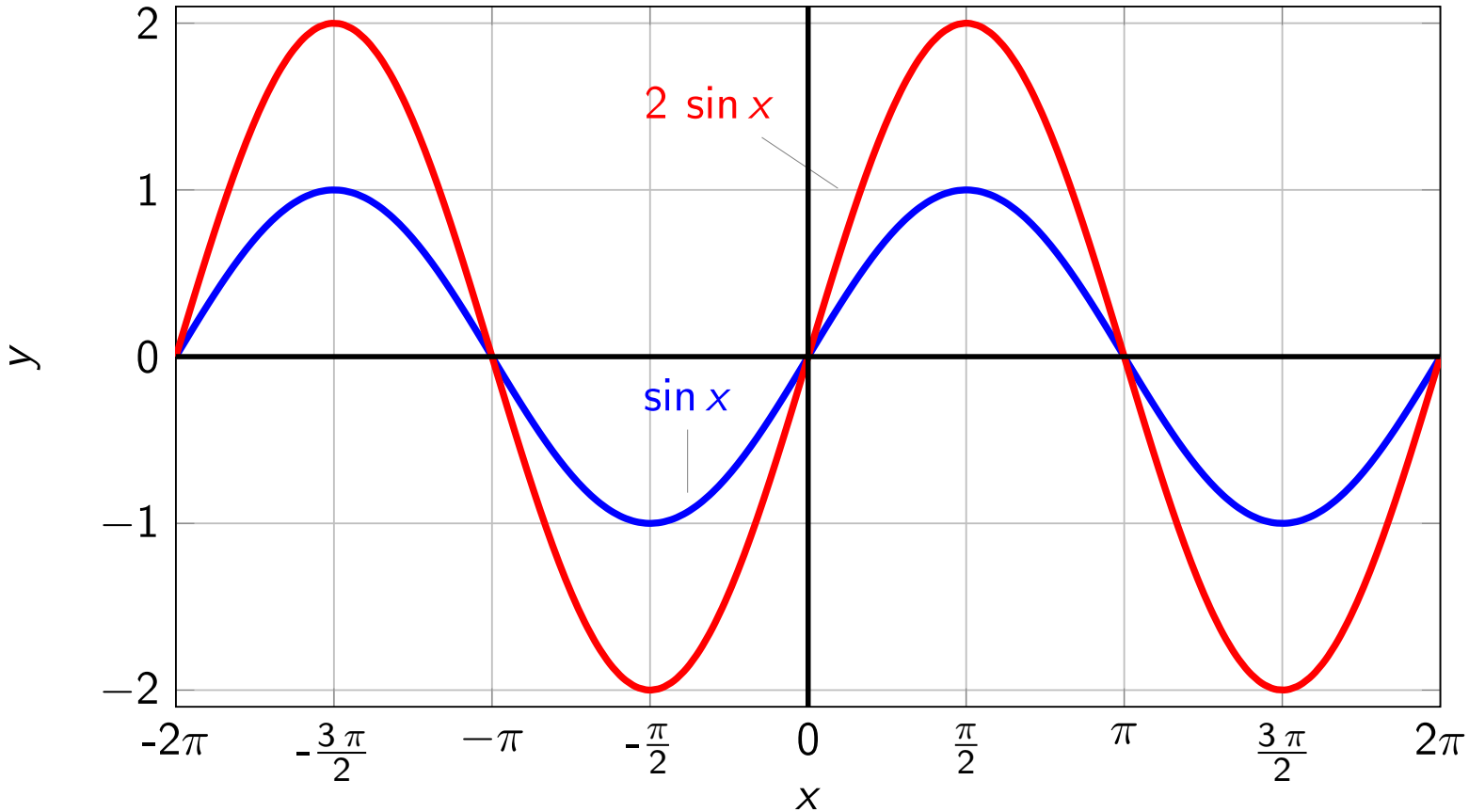
- The graph of $y_3 = k\, f(x)$ is obtained from that of f by multiplying all the ordinates $f(x)$ by k.
- In particular, if $k = -1$ the ordinates simply change sign, so that the graph of y_3 is symmetric to that of f with respect to the x-axis.
- Note that if $k > 1$, the graph is "**stretched**" in the **vertical direction**, dilating the **positive** ordinates **upward** and the **negative** ones **downward**.

Vertical dilation and contraction (2/2)

- Conversely, if $0 < k < 1$ the graph "**shrinks**", again in the vertical direction.
- Therefore, the operation of multiplying $f(x)$ by k has the geometric meaning of a **dilation** (if $|k| > 1$) or a **contraction** (if $|k| < 1$) along the y-axis, possibly accompanied by a **reflection with respect to the x-axis**, if $k < 0$.

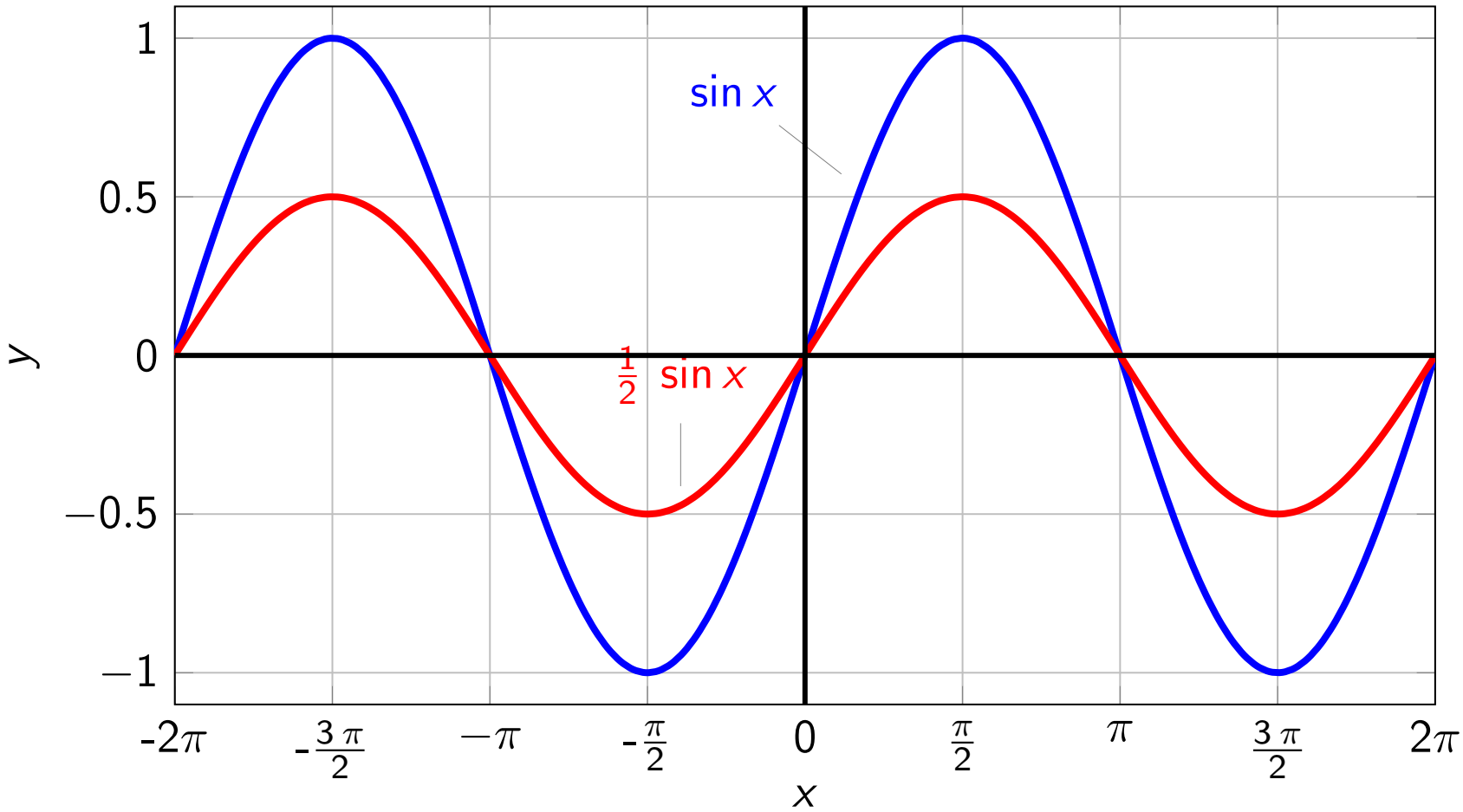
Example (1/3)

Consider $y = \sin x$, then $y_3 = k \sin(x)$. With $k = 2$:



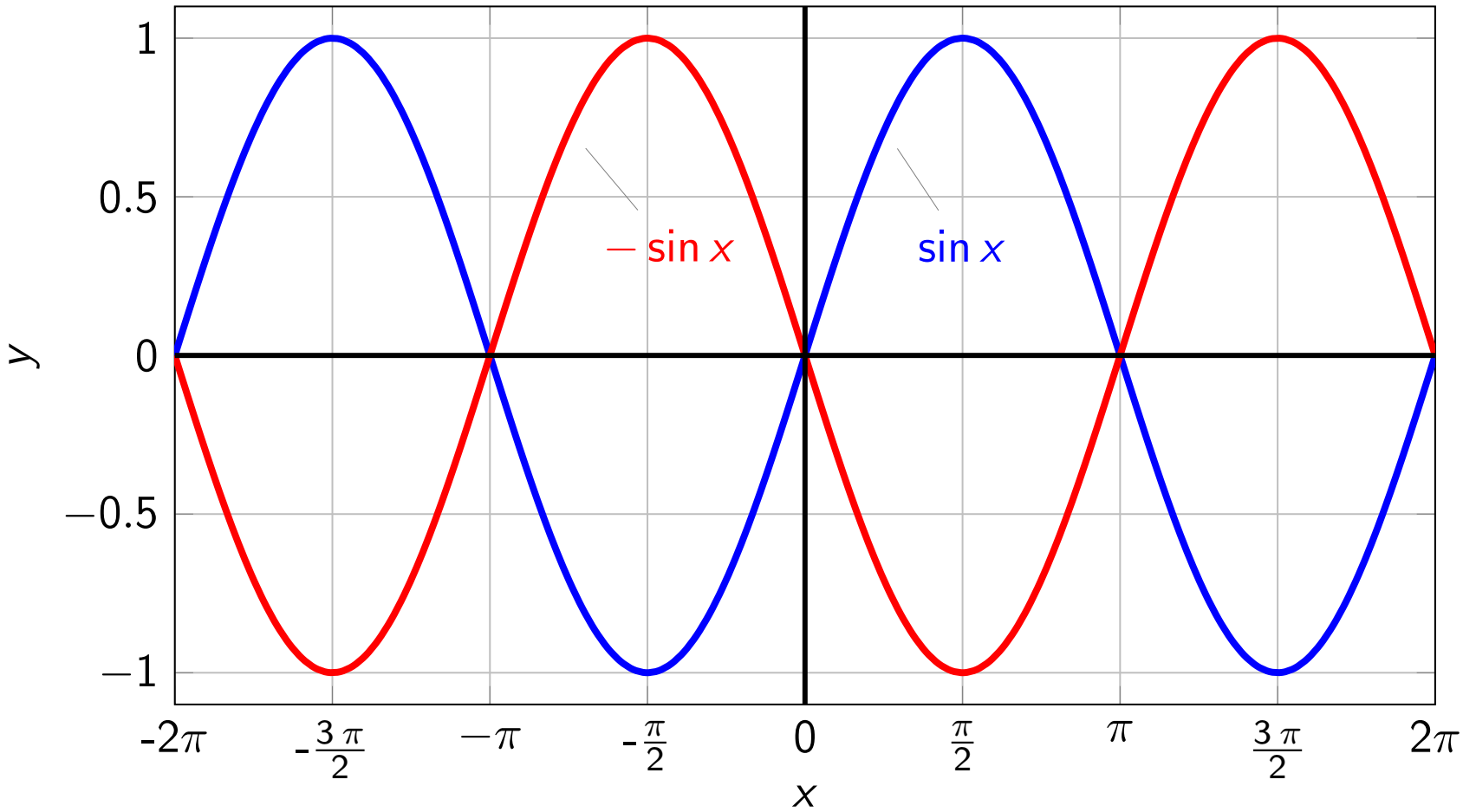
Example (2/3)

With $k = \frac{1}{2}$:



Example (3/3)

With $k = -1$:



Summary

$y_3 = k\,f(x)$:

- All the ordinates are multiplied by k
- $|k| > 1$: vertical dilation; $|k| < 1$: vertical contraction
- $k < 0$: also a reflection with respect to the x-axis (for $k = -1$ only a reflection)

Operations related to $y_4 = f(k\,x)$

Introduction: Operations related to $y_4 = f(k\,x)$

- We see what happens to the graph when we **multiply the variable** x by a constant k, before applying f
- We expect a **horizontal change of scale**: the graph is compressed or dilated along the x-axis, and possibly reflected

Horizontal change of scale (1/2)

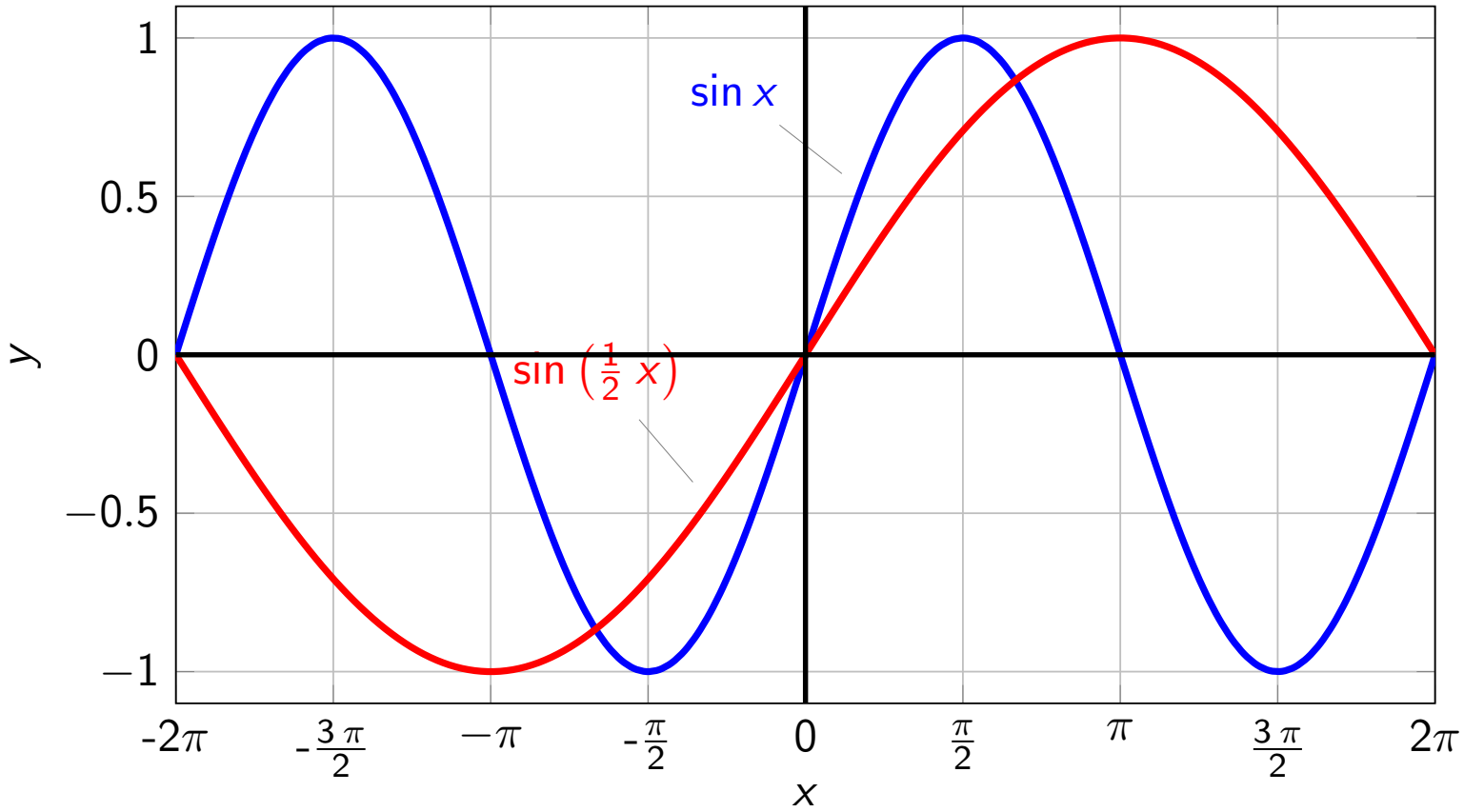
- The graph of $y_4 = f(k\,x)$ is obtained from that of $f(x)$ by a **change of scale** along the x-axis.
- If $k > 1$, $k\,x$ grows faster than x, and therefore the graph of y_4 will be similar to that of f but with faster oscillations, that is, it will be "**compressed**" in the **horizontal direction** by a factor $\frac{1}{k}$.

Horizontal change of scale (2/2)

- Similarly, if $0 < k < 1$ the graph will appear "**dilated**" in the **horizontal direction**, with gentler oscillations.
- If $k < 0$, in addition to a compression (if $|k| > 1$) or a dilation (if $|k| < 1$) along the x-axis, there will be a **reflection with respect to the y-axis**.

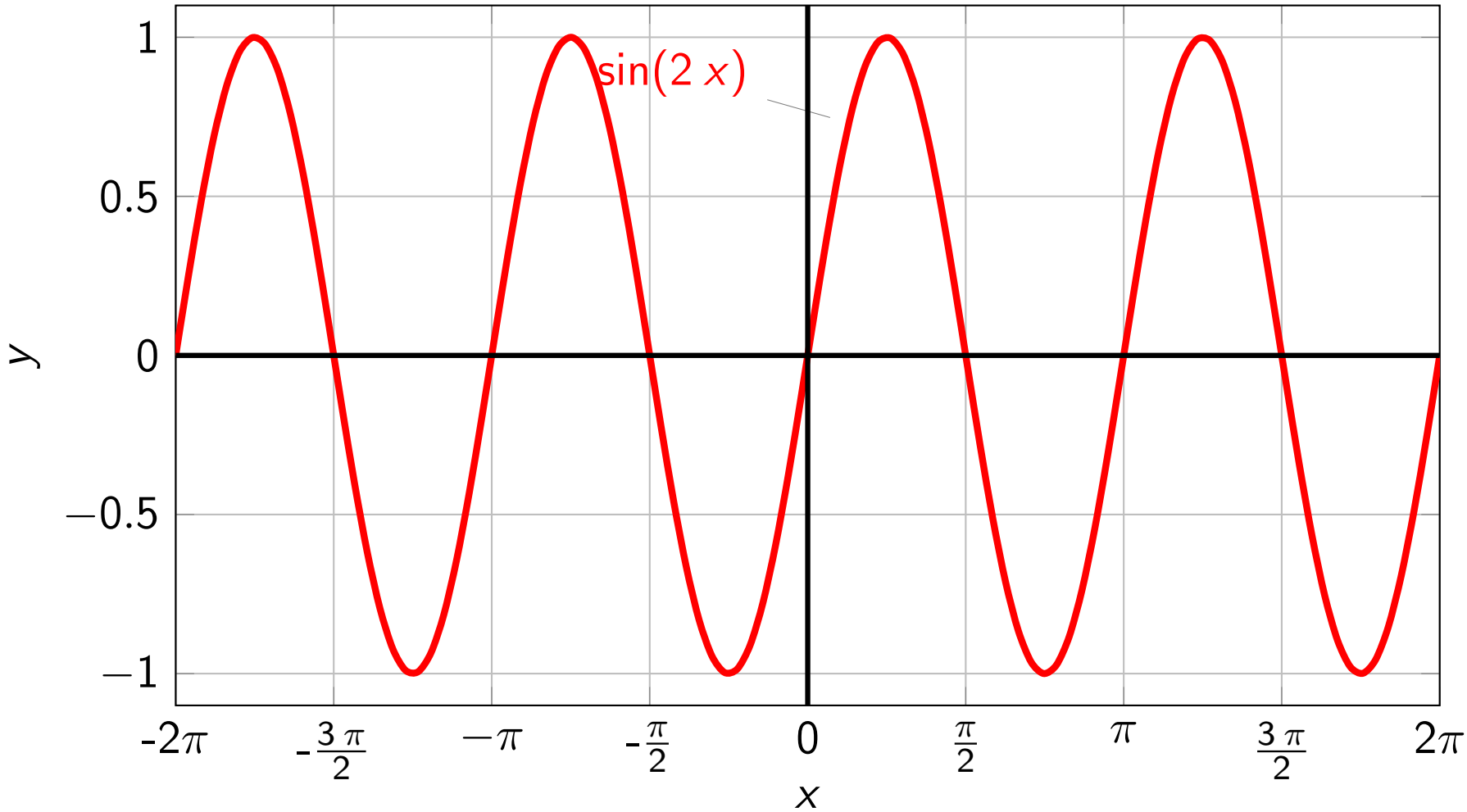
Example (1/2)

Consider $y = \sin x$, then $y_4 = \sin(k\,x)$. With $k = \frac{1}{2}$ and $k = 2$ we have:



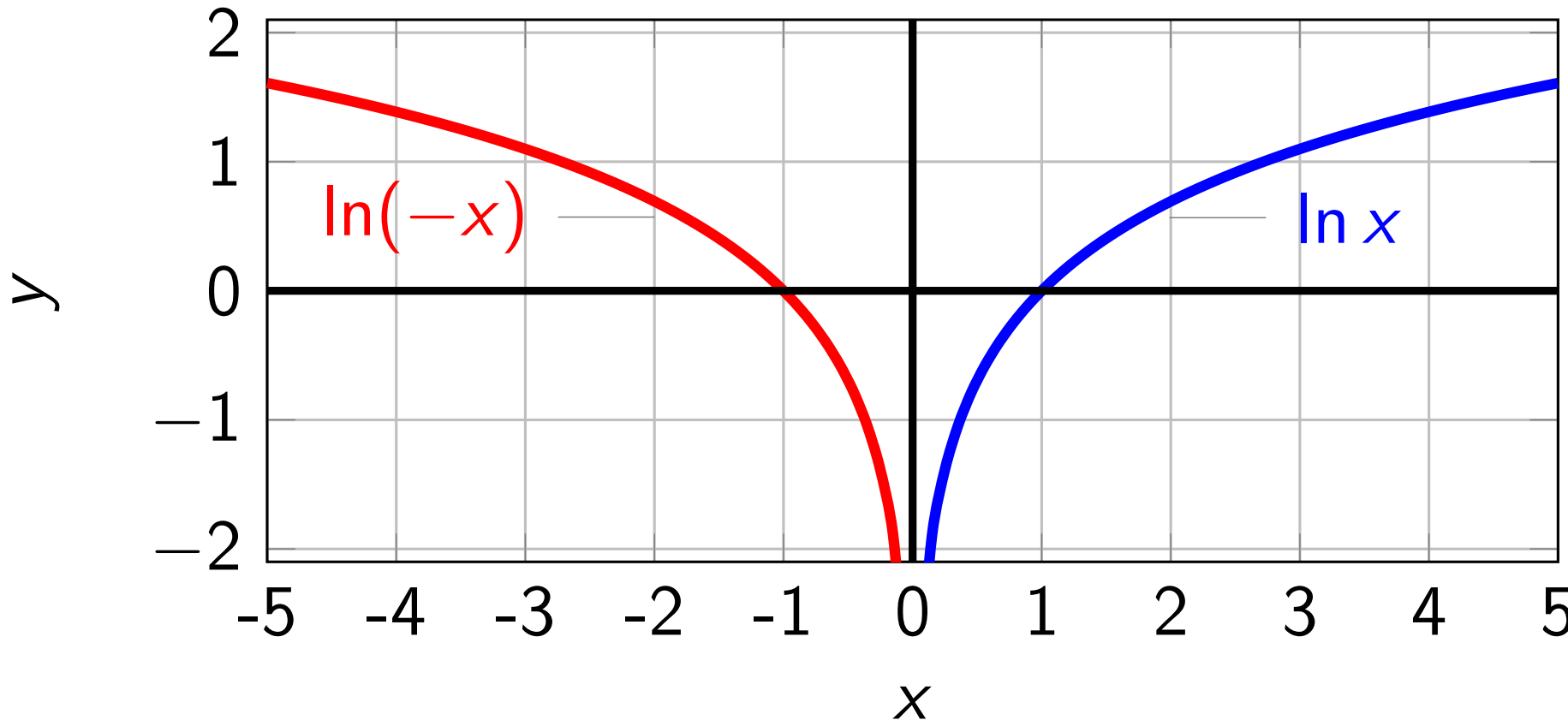
Example (2/2)

With $k = 2$:



Example

Consider $y = \ln x$, then $y_4 = \ln(k\,x)$. With $k = -1$ we have:



Summary

$y_4 = f(kx)$:

- Change of scale along the x-axis
- $|k| > 1$: horizontal compression by a factor $\frac{1}{|k|}$; $|k| < 1$: horizontal dilation
- $k < 0$: also a reflection with respect to the y-axis (e.g. $\ln(-x)$)

Operations related to $y_5 = |f(x)|$

Introduction: Operations related to $y_5 = |f(x)|$

- We see how to obtain the graph of the **absolute value** of a function
- The absolute value makes all values non-negative: the part of the graph below the x-axis is "flipped" upward

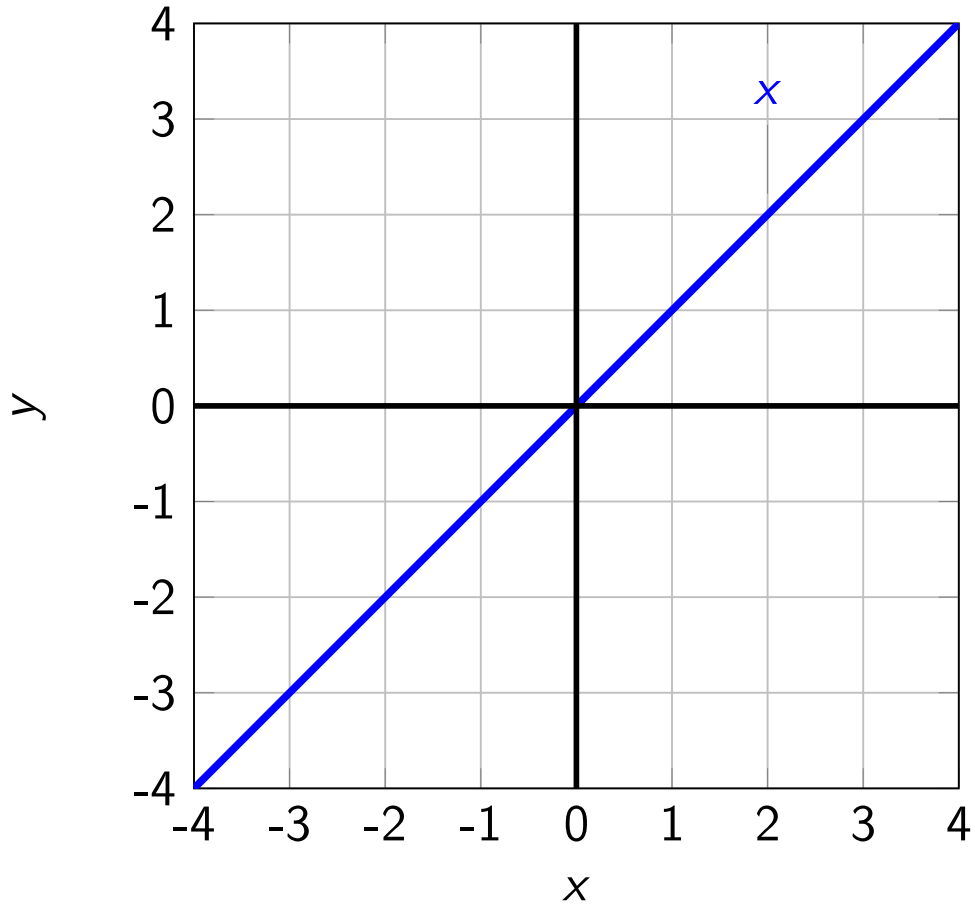
Absolute value of $f(x)$

$$|f(x)| = \begin{cases} f(x) & \text{if } f(x) \geq 0, \\ -f(x) & \text{if } f(x) < 0 \end{cases}$$

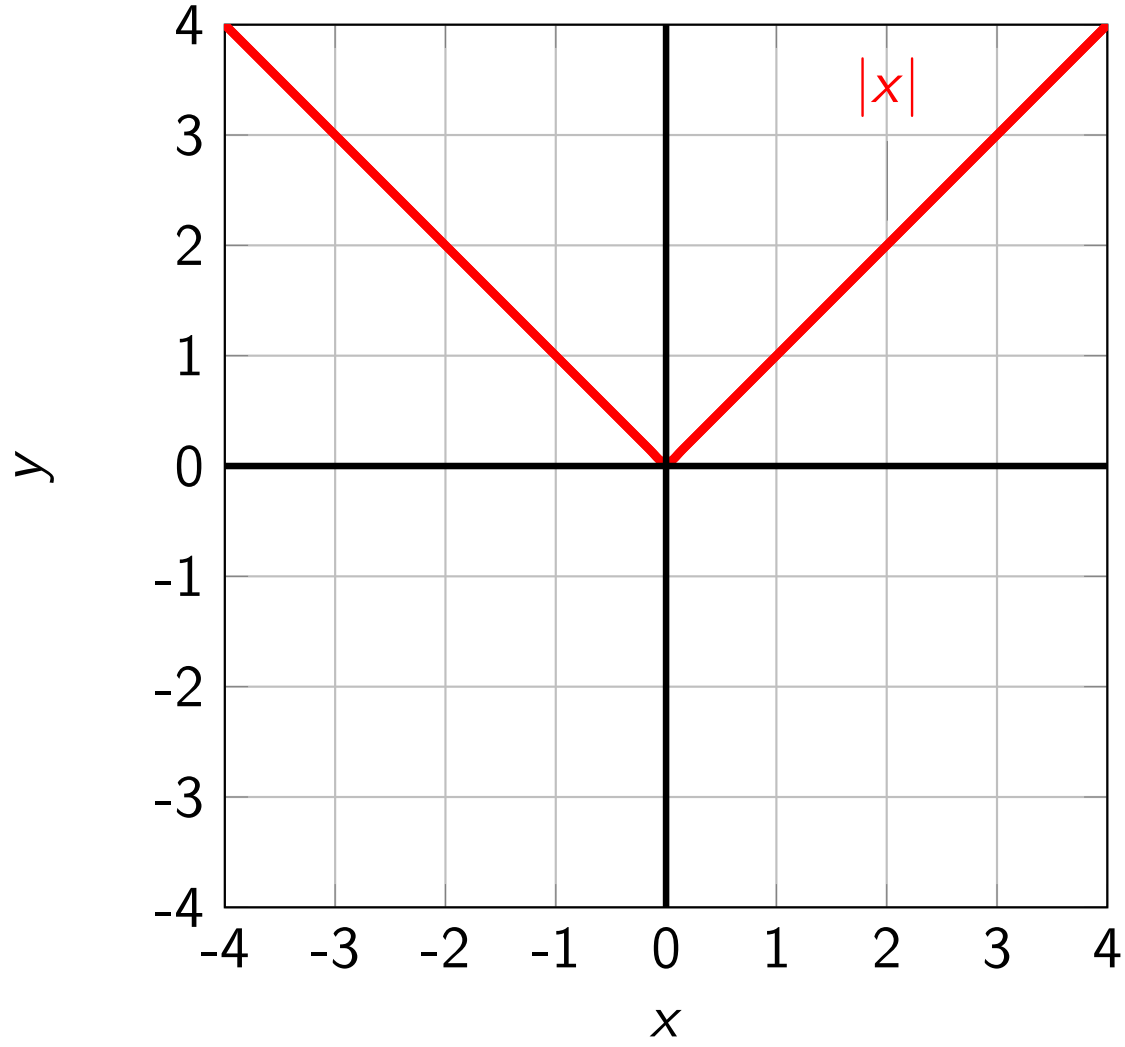
- When passing from the graph of $y = f(x)$ to that of $y_5 = |f(x)|$, the points with **non-negative ordinate remain unchanged**, while those with **negative ordinate are transformed into their symmetric points with respect to the x-axis**.
- The graph of $y_5 = |f(x)|$ is obtained from that of f by "flipping" symmetrically with respect to the x-axis the part of the graph of f that lies in the lower half-plane, and leaving the rest unchanged.

Example (1/2)

Consider $y = x$, then $y_5 = |x|$

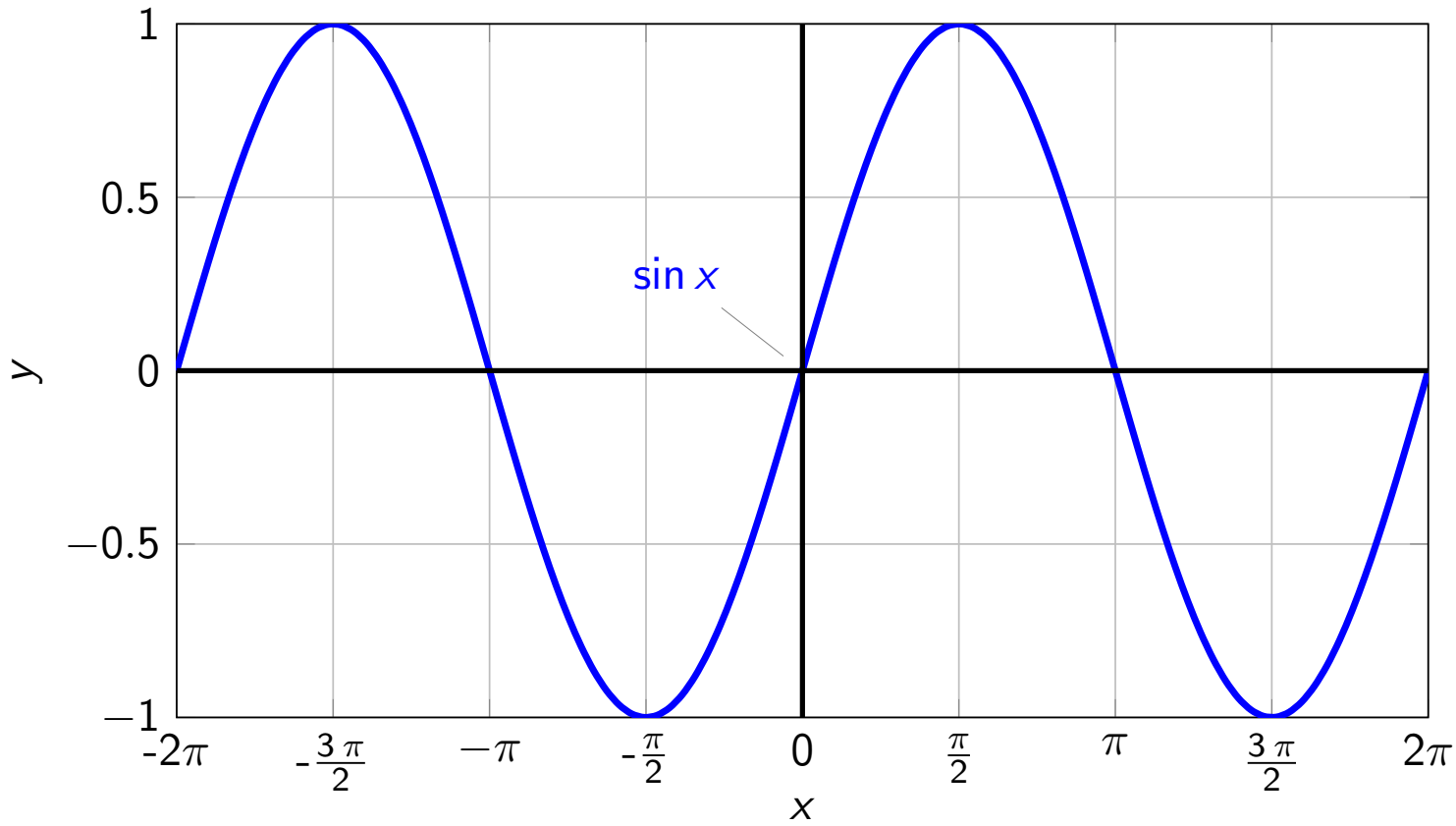


Example (2/2)

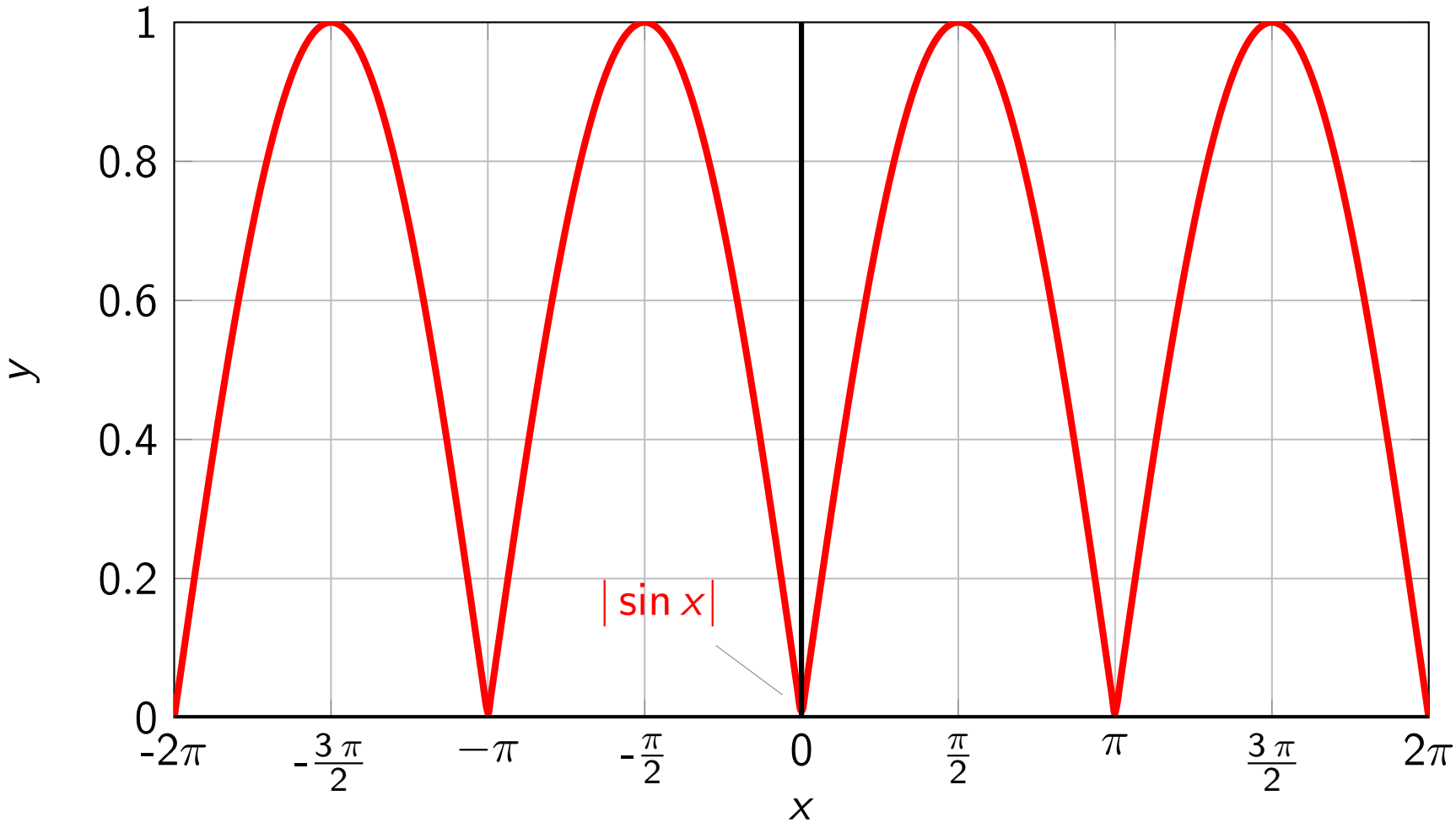


Example (1/2)

Consider $y = \sin x$, then $y_5 = |\sin x|$



Example (2/2)



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Summary

$y_5 = |f(x)|$:

- The points with non-negative ordinate remain unchanged
- The part of the graph in the lower half-plane is flipped with respect to the x-axis
- The graph of $|f(x)|$ lies entirely in the half-plane $y \geq 0$

Operations related to $y_6 = f(|x|)$

Introduction: Operations related to $y_6 = f(|x|)$

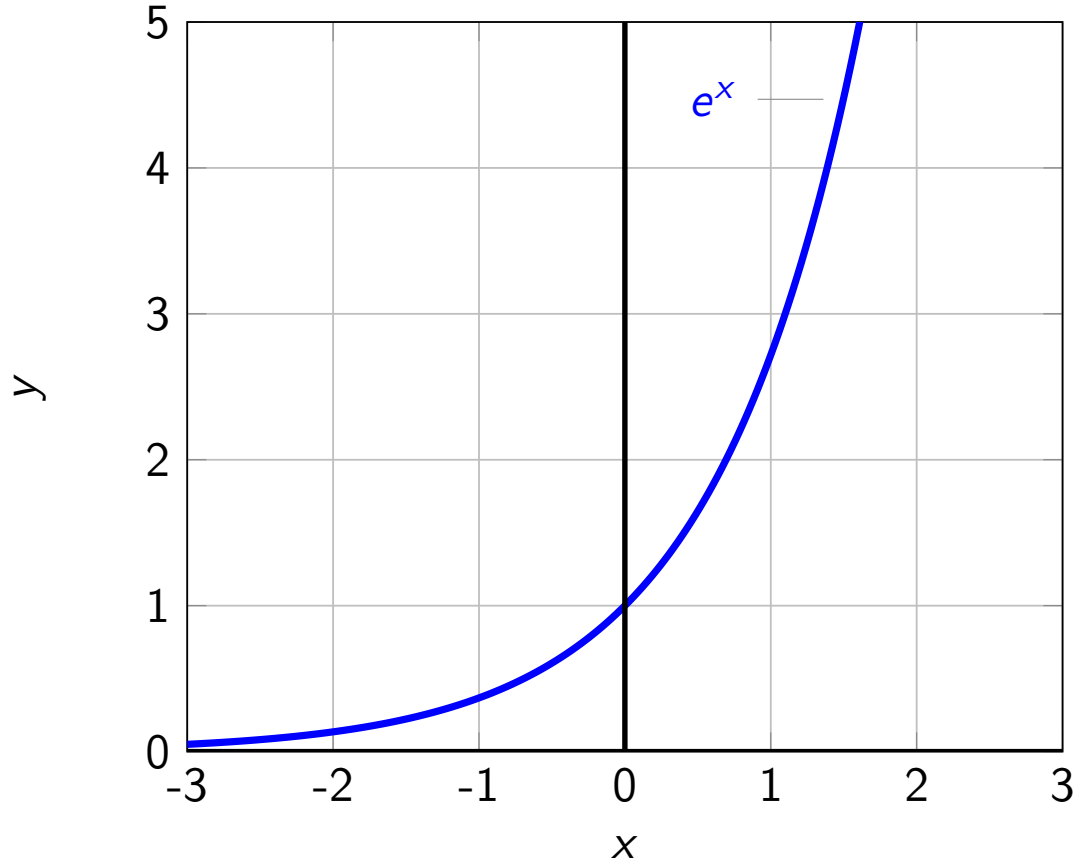
- Finally, we see the case in which the absolute value is applied to the **variable** x
- We always obtain an **even** function, whose graph is symmetric with respect to the y-axis

Graph of $y_6 = f(|x|)$

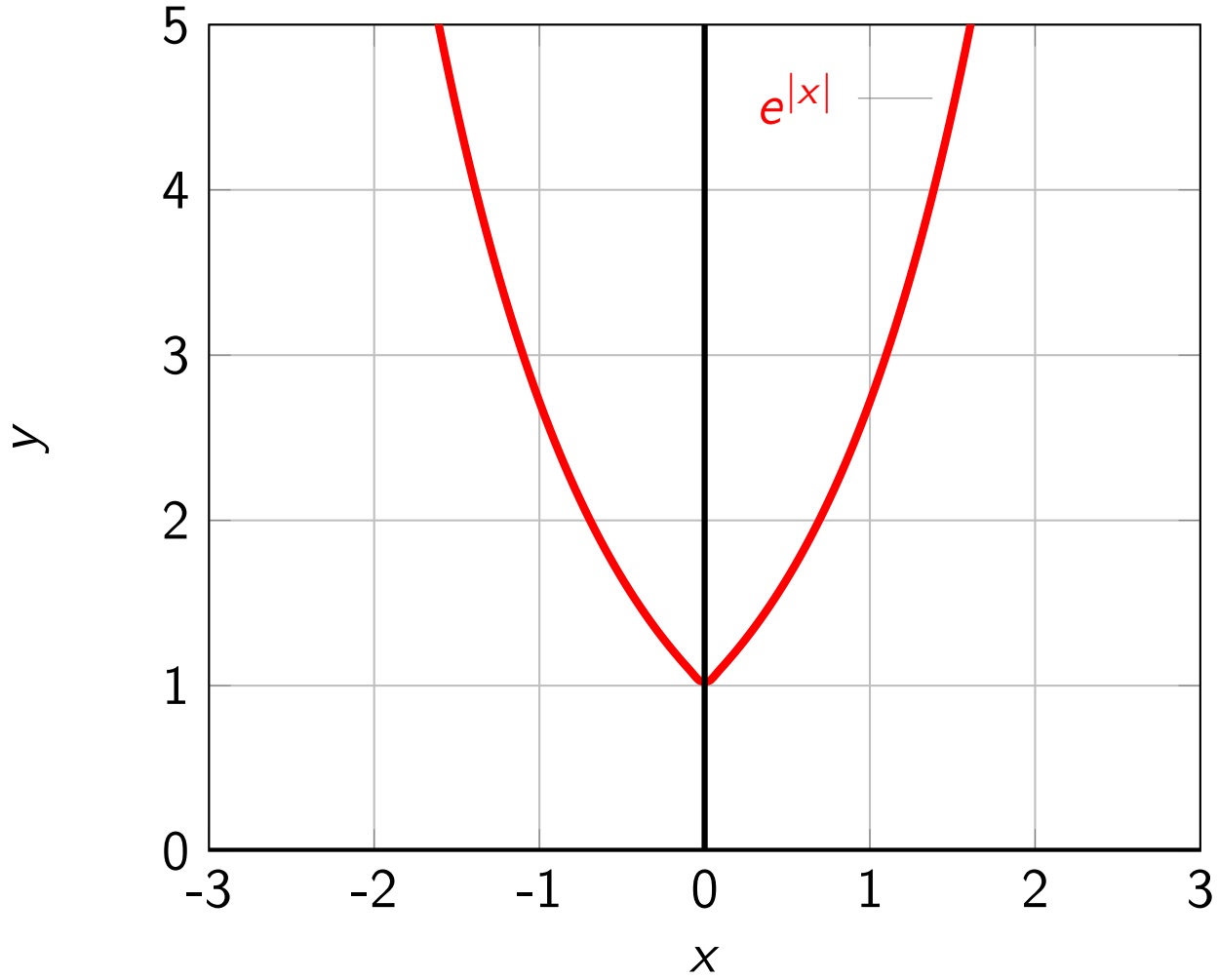
- Finally, to draw the graph of $y_6 = f(|x|)$, we observe that $|x| = x$ for $x \geq 0$, and hence in the right half-plane the two graphs coincide.
- We have $|-x| = |x|$, and hence y_6 is an even function, therefore symmetric with respect to the y-axis.
- Consequently, the graph of $y_6 = f(|x|)$ is drawn by **leaving the graph of f unchanged in the right half-plane and flipping it symmetrically with respect to the y-axis**.

Example (1/2)

Consider $y = e^x$, then $y_6 = e^{|x|}$



Example (2/2)



Summary

$y_6 = f(|x|)$:

- In the right half-plane ($x \geq 0$) the graph coincides with that of f
- $f(|x|)$ is even: the right part of the graph is flipped with respect to the y-axis
- Example: $e^{|x|}$

Summary (1/2)

Translations:

- $f(x) + a$: vertical translation (upward if $a > 0$, downward if $a < 0$)
- $f(x + a)$: horizontal translation (to the left if $a > 0$, to the right if $a < 0$)

Dilations and reflections:

- $k\, f(x)$: vertical dilation/contraction, reflection with respect to the x-axis if $k < 0$
- $f(kx)$: horizontal compression/dilation, reflection with respect to the y-axis if $k < 0$

Summary (2/2)

Absolute values:

- $|f(x)|$: the part of the graph below the x-axis is flipped with respect to the x-axis
- $f(|x|)$: the part for $x \geq 0$ is kept and flipped with respect to the y-axis

- Combining these operations, we can draw the graphs of many functions starting from those of the elementary functions